

Question Paper

Exam Date & Time: 05-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026
SOFTWARE ENGINEERING [DS_3152]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) How can the three fundamental types of program constructs be differentiated based on their purpose and use in solving a problem? Illustrate each construct with suitable examples from a high-level programming language of your choice. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Using the given program that employs **goto** statements for login verification, demonstrate how the same logic can be implemented using structured programming. (5)

```
void verify_login(char username[], char password[]) {
    int valid_user = 0;

    if (strcmp(username, "admin") == 0 && strcmp(password,
"1234") == 0) {
100:  valid_user = 1;
        goto step110;
    }
    else if (strcmp(username, "guest") == 0)
        goto step100;    // allow guest login
    else
        goto step120;    // invalid credentials

step110:
    if (valid_user == 1)
        printf("Access Granted. Welcome, %s!\n", username);
        goto step130;

step120:
    printf("Error: Invalid Username or Password.\n");

step130:
    printf("Login process completed.\n");
}
```

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Apply your understanding of requirements engineering to identify and describe various requirements gathering techniques commonly used by analysts, explaining how each technique is implemented in

A) practice.

Course Outcome: "CO2" / Cognitive Level: "Application"

B) Illustrate the Rapid Application Development (RAD) life cycle model and demonstrate how it enables faster software development while effectively handling user change requests. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

3) How can the concept of functional independence be described in the context of software design? (5)
Discuss how design principles such as cohesion and coupling contribute to achieving functional independence, and explain the advantages this provides in developing reliable and maintainable software

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) Explain the concept of cohesion in software design. Using suitable examples, describe the following types of cohesion: (5)

1. Logical cohesion
2. Procedural cohesion

Course Outcome: "CO3" / Cognitive Level: "Application"

4) An Insurance Management System assists a company in handling claim requests efficiently. The system performs the following key operations: (5)

A)

1. Policyholder Registration
2. Claim Submission
3. Claim Verification
4. Decision on Approval or Rejection
5. Claim Status Update and Customer Notification

Using the principles of UML Communication Modelling, illustrate the interactions among major entities such as *Customer*, *Claim Handler*, *Verification Unit*, *Database*, and *Notification Module* using sequence diagram.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) A Library Management System is designed to automate various library operations such as member registration, book issuance, returns, fine calculation, and inventory updates. The major functions of the system include: registering members, searching and issuing books, returning books, calculating fines for late submissions, and updating the book inventory. Using the concepts of UML Component Diagrams, demonstrate how these operations can be organized into modular software components. Identify and illustrate the interaction among key components such as the *User Interface*, *Book Database*, *Transaction Manager*, *Fine Calculator*, and *Report Generator*. Draw a schematic Component Diagram showing the interfaces, dependencies, and relationships among these components to represent the functional architecture of the system. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) Design a black-box test suite for a function that takes the name of a book as input and searches a file containing the names of the books available in the Library and displays the details of the book if the book is available in the library otherwise displays the message "book not available". (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Apply your understanding of integration testing to illustrate how stubs and drivers are used in the (5)

process, and demonstrate their role in facilitating effective integration testing.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	DS_3152:SOFTWARE ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	DS_3152:SOFTWARE ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)